

SUPUNG, Korea, Democratic People's Republic of

WMO: 470280

Lat: 40.450N

Lon: 124.933E

Elev: 249

StdP: 14.56

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	1.0	4.2	-10.2	3.2	7.3	-7.0	3.8	8.6	13.9	14.8	12.2	15.4	3.6	180	0.376

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	88.6	75.5	86.0	74.1	83.8	73.2	79.0	85.0	77.2	82.4	75.8	80.5	2.7	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.2	142.9	83.1	75.6	135.3	80.5	74.4	129.6	78.8	42.8	85.5	41.0	82.4	39.6	80.8	87.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.9	9.8	8.1	-4.0	94.0	4.3	2.7	-7.1	95.9	-9.6	97.5	-12.0	99.0	-15.1	101.0
			-4.3	81.3	4.5	2.5	-7.5	83.1	-10.1	84.5	-12.6	85.9	-15.8	87.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	49.5	18.9	24.1	36.3	50.0	61.5	70.3	76.1	75.4	66.0	53.1	37.5	22.7	(6)
(7)		DBStd	21.04	6.97	7.33	6.78	6.77	5.14	3.72	3.41	4.11	5.10	6.40	8.54	7.84	(7)
(8)		HDD50	3466	963	725	427	82	1	0	0	0	0	42	381	845	(8)
(9)		HDD65	6604	1428	1145	890	451	132	5	0	1	48	370	824	1310	(9)
(10)		CDD50	3267	0	0	2	80	357	608	808	788	479	138	7	0	(10)
(11)		CDD65	931	0	0	0	0	23	163	343	324	77	1	0	0	(11)
(12)		CDH74	6342	0	0	0	15	322	1093	2307	2205	395	5	0	0	(12)
(13)		CDH80	1649	0	0	0	0	45	255	670	640	39	0	0	0	(13)
(14)	Wind	WSAvg	2.6	3.1	2.6	2.7	2.7	2.4	2.0	1.8	1.9	2.5	3.1	3.4	3.3	(14)
(15)	Precipitation	PrecAvg	42.0	0.3	0.7	0.9	2.2	3.1	4.9	11.6	10.1	3.6	2.2	1.7	0.7	(15)
(16)		PrecMax	70.7	1.1	2.8	2.8	3.9	7.5	9.8	26.9	30.9	11.8	6.4	3.9	2.1	(16)
(17)		PrecMin	27.4	0.0	0.0	0.0	0.0	0.6	0.9	5.2	1.5	0.6	0.1	0.3	0.1	(17)
(18)		PrecStd	10.0	0.3	0.7	0.6	1.0	1.6	2.3	4.9	6.1	2.4	1.5	1.1	0.5	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	39.7	46.5	62.2	76.1	84.9	88.8	93.0	92.0	83.7	73.9	60.8	44.9	(19)
(20)			MCWB	35.3	39.2	47.6	56.6	65.7	69.8	78.4	79.9	71.6	62.3	53.5	41.2	(20)
(21)		2%	DB	36.1	41.9	56.2	71.8	80.7	85.7	89.1	88.2	80.7	69.8	57.1	40.5	(21)
(22)			MCWB	32.5	35.1	45.4	55.8	63.5	69.2	76.5	77.1	69.3	59.2	50.4	37.1	(22)
(23)		5%	DB	33.3	38.7	52.0	67.6	77.7	83.0	86.1	85.7	78.4	66.9	53.4	37.1	(23)
(24)			MCWB	30.0	33.7	42.7	52.8	62.0	68.7	75.4	75.4	67.0	57.5	47.8	34.0	(24)
(25)		10%	DB	30.2	36.0	48.3	63.7	74.6	80.4	83.6	83.5	76.1	63.9	50.1	34.2	(25)
(26)			MCWB	27.0	31.3	40.5	50.8	60.7	68.1	74.9	74.4	65.8	55.8	45.4	31.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	36.3	40.3	50.5	60.7	69.2	74.3	82.4	81.7	74.1	63.9	54.9	42.6	(27)
(28)			MCDB	38.5	44.8	57.4	69.4	80.4	83.2	90.4	89.6	80.9	70.6	58.7	44.6	(28)
(29)		2%	WB	33.2	36.7	46.9	57.5	66.2	72.5	79.5	79.2	71.5	61.2	51.7	37.6	(29)
(30)			MCDB	35.8	40.3	54.7	67.7	76.2	80.6	85.4	85.7	78.0	67.6	56.0	40.0	(30)
(31)		5%	WB	30.0	34.2	43.9	54.8	64.2	71.0	77.4	77.6	69.6	59.2	48.8	34.4	(31)
(32)			MCDB	32.8	38.0	50.5	65.3	74.3	78.9	82.6	83.1	75.5	64.5	52.6	36.5	(32)
(33)		10%	WB	27.2	31.7	41.2	52.0	62.2	69.8	76.0	76.2	68.0	57.4	45.7	31.7	(33)
(34)			MCDB	30.1	35.7	47.6	61.4	71.5	77.4	80.9	81.0	73.2	62.5	49.4	34.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	16.7	17.5	18.4	21.8	22.2	18.1	13.6	14.7	18.7	19.0	14.7	14.2	(35)
(36)			MCDBR	17.6	19.9	24.3	28.3	28.4	24.4	18.9	18.4	21.0	21.8	18.7	14.8	(36)
(37)			MCWBR	14.3	14.6	15.0	14.3	12.9	10.2	8.2	8.1	9.9	12.2	13.2	12.4	(37)
(38)		5% WB	MCDBR	16.9	17.9	22.7	25.7	24.2	18.2	14.8	14.4	16.6	19.0	16.1	14.1	(38)
(39)			MCWBR	14.2	14.2	15.2	14.3	12.5	9.0	7.7	7.5	8.9	11.9	12.4	12.5	(39)
(40)																
(41)	Clear-Sky Solar Irradiance	taub	0.392	0.473	0.539	0.626	0.656	0.731	0.698	0.596	0.523	0.551	0.479	0.405	(40)	
(42)		taud	1.944	1.792	1.719	1.609	1.591	1.524	1.633	1.834	1.938	1.774	1.863	1.946	(41)	
(43)		Ebn at Noon	225	222	223	214	213	197	202	220	226	196	192	207	(42)	
		Edh at Noon	41	56	67	80	83	89	80	63	54	56	45	38	(43)	
(44)	All-Sky Solar Radiation	RadAvg	781	1019	1360	1545	1747	1561	1308	1400	1384	1039	717	650	(44)	
(45)		RadStd	39	65	94	112	135	180	148	168	116	66	80	49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (15 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page